

Lorenzo Vianello

Robotics Researcher · Postdoctoral Fellow · Shirley Ryan AbilityLab

Chicago, IL

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Lorenzo Vianello works at the intersection of robotics, control theory, and clinical rehabilitation science, developing exoskeleton and prosthetic control systems for stroke and mobility rehabilitation. He is co-first author of the flagship Therapist-Exoskeleton-Patient Interaction (TEPI) study published in *Science Robotics* (2026), recipient of the IEEE Best Paper Award in Medical Robotics at ICRA 2024, and author of 26 peer-reviewed publications (229 citations, h-index 7, i10-index 6, Google Scholar, as of August 17, 2026).

Experience

Center for Bionic Medicine, Shirley Ryan Ability Lab / Northwestern University

Chicago, IL

Postdoctoral fellow: Control and design of Lower-limb Prosthetic device and Exoskeleton

August 2025 -

- Developed Controllers for Prosthetic Devices and Exoskeletons.

Related skills: Prosthesis, Exoskeleton, Controls, MicroPython, Embedded systems, Medical Robots

Legs+Walking Lab, Shirley Ryan Ability Lab / Northwestern University

Chicago, IL

Postdoctoral fellow: Design Control Strategies for Human-Exoskeleton Interaction

March 2023 - August 2025

- Developed bilateral teleoperation platform for Robot-mediated physical Interaction. ([Paper Accepted at Science Robotics](#))
- Developed whole-body controllers for exoskeleton imposing safety and balancing constraints ([Paper](#))
- Developed Machine-Learning based Exoskeleton Controllers([Paper](#)).
- Supervised four PhD and six Master's students.

Related skills: Machine Learning, Legged Robots, Exoskeletons, teleoperation, controls, C++, Python, Matlab, ROS

CRAN and LORIA Nancy

Nancy, FR

PhD Thesis: Toward adaptation in human-robot collaboration ([Thesis](#))

December 2019 - December 2022

- Investigated Control algorithms for robots manipulators physically interacting with humans ([Paper](#)).
- Developed a real-time human intention detection and Digital Human Model simulated in RobotDART ([Paper](#)).
- Developed a posture prediction algorithm for human interacting with robot manipulators ([Paper](#)).

Related skills: Machine Learning, Robot manipulator, Controls, Simulation, C++, Python, Matlab, ROS

TU-Delft

Delft, NL

Visiting Scholar at Delft University of Technology

February 2022 - June 2022

- Developed a tele-impedance based collaborative robot controller ([Paper](#)).
- Designed Controllers for high-DoF robot manipulator, implemented safety constraints.

Related skills: Tele-impedance, force/motion control, pHRI, C++, ROS, Python, EMG

LORIA Nancy

Nancy, FR

Research Intern / Master Thesis: Human Guided Grasp Planning ([Thesis](#))

March 2019 - Sept 2019

- Developed an Autonomous grasp planning algorithm that learns from user demonstrations and plans motion in 3D space.

Related skills: Computer Vision, Auto-Encoders, Gaussian Processes, Manipulator Control (Moveit)

Sapienza University

Rome, IT

Graduation Project: Torso Pose Estimation on the HRP4 Simulated Robot([Project](#))

Sept 2017- Dec 2017

- Implemented an Extended Kalman Filter to localize robot pose from multiple sensors.

Related skills: Humanoid whole-body control, Extended-Kalman-Filter, C++, V-REP Simulation

Education

University of Lorraine

Nancy, France

Phd in Human-Robot mutual collaboration

December 2022

- Advisers: Serena Ivaldi and Alexis Aubry
- Dissertation Title: Toward adaptation in human-robot collaboration
- Acquired skills: Robotics, Controls, Machine Learning, Simulation, System Optimization (TA)

- Thesis Advisers: Giuseppe Oriolo
- GPA: 110/110 - Student Honor Program
- Relevant Courses: Robotics, Probabilistic Robotics, Machine Learning, Artificial Intelligence, Vision and Perception

Padua University

BS in Information Engineering

*Padua, Italy**Sept 2017*

- Relevant Courses: Automatic Controls, Electronics, Telecommunications, Informatics, Physics and Mathematics

Skills

Robotics Fundamentals, High DoF systems, Motion/Force Controllers, Simulations, Solvers, ROS, ROS2, Gazebo, RobotDART, V-Rep

Machine Learning Tensorflow, TF-lite, Keras, PyTorch

Programming C++, Python, Matlab/Simulink, Git, Docker

Awards & Scholarships

2025	2nd Place Post Doctoral Research Fellow & Graduate Student Sarah Baskin Award,	<i>Chicago</i>
2024	Best Paper Award in Medical Robotics and Finalist in Best Paper Award, ICRA 2024	<i>Japan</i>
2023	Finalist for Best PhD thesis 2023 by GDR MACS, INRIA Nancy	<i>France</i>
2020	Finalist Best Interactive Paper Award, Humanoids 2020	<i>Remote</i>
2019	Student Honors Program, Sapienza University	<i>Rome, Italy</i>
2019	Bordoni Scholarship, Sapienza University	<i>Rome, Italy</i>

Media Exposure & Public Recognition

- **Crain's Chicago Business** – “Chicago researchers test connected exoskeletons to help stroke patients walk,” June 19, 2026. Names, pictures, and directly quotes Dr. Vianello as co-first author.
- **WGN-TV** (Chicago, Channel 9) – “Exoskeletons revolutionize physical therapy for stroke,” TV segment, June 23, 2026.
- **Fox 32 Chicago** – “Innovative exoskeleton therapy for stroke survivors developed in Chicago,” TV segment, June 20, 2026.
- **Yahoo News** – national syndication of the WGN-TV segment, June 23, 2026.
- **Northwestern Now** – “Robotic exoskeleton could redefine how stroke survivors relearn to walk,” June 17, 2026.
- **Shirley Ryan AbilityLab News** – “First-of-Its-Kind Exoskeleton Therapy Could Redefine How Stroke Survivors Relearn to Walk,” June 16, 2026.
- **NR Times** (UK) – “Linked robotic exoskeletons could enhance gait retraining after stroke,” June 23, 2026. Names and quotes Dr. Vianello directly.
- **Healthcare in Europe** – “Exoskeleton therapy could help stroke survivors relearn to walk,” June 18, 2026. Quotes Dr. Vianello directly.
- **Medical Xpress** – “First-of-its-kind exoskeleton therapy could redefine how stroke survivors relearn to walk,” June 17, 2026. Quotes Dr. Vianello by name and credential.
- **Additional international pickup** – Institution of Mechanical Engineers (IMechE, UK), Mirage News (Australia), ConnectSci (Australia), Bioengineer.org, News-Medical.Net, Medical Design & Development, and Hoodline.
- **IEEE Robotics & Automation Magazine** (official IEEE RAS publication) – “RAS Congratulates the Honorees at the ICRA 2024 Awards Ceremony,” September 2024 issue. Names Dr. Vianello as first author of the Best Paper Award in Medical Robotics.
- **Northwestern Center for Robotics and Biosystems News** – “SRAlab-CRB Collaboration Wins Best Medical Robotics Paper Award at ICRA 2024,” May 20, 2024. Names Dr. Vianello as co-first author and pictures him receiving the award in Yokohama, Japan.

Peer-Review & Editorial Service

- **IEEE Robotics and Automation Letters** – 9 manuscripts (per journal reviewer portal).
- **IEEE Transactions on Medical Robotics and Bionics** – 4 manuscripts across two review cycles.

- **Scientific Reports** – 2 manuscripts (Springer Nature Reviewer Certificates issued for both).
- **IEEE Transactions on Robotics** – 1 manuscript (per journal reviewer portal).
- **IEEE/ASME Transactions on Mechatronics, Journal of Field Robotics, Applied Sciences** (MDPI), and **Disability and Rehabilitation: Assistive Technology** – 1 manuscript each.
- Conference reviewing: IEEE/RSJ IROS, with additional review invitations across ICRA, ICORR, RO-MAN, BioRob, and ICNR.
- **Guest Editor**, Special Issue, *Journal of NeuroEngineering and Rehabilitation* – 2025.
- **Guest Editor**, Special Issue, *Journal of Wearable Technologies* – 2025.

Journal Publications (Full Author Lists)

- Küçüktağak E. B., Short M. R., **Vianello, L.**, Ludvig D., Hargrove L., Lynch K., Pons J. L. *Therapist-Exoskeleton-Patient Interaction for Gait Therapy*. Science Robotics, 2026.
- Short M. R., Bandini L., Ludvig D., **Vianello, L.**, Sanguineti V., Pons J. L. *Haptic Interaction with a Human Partner for Ankle Training in Chronic Stroke: A Pilot Study*. Journal of NeuroEngineering and Rehabilitation, 2026.
- Lhoste C., Cantón A., Küçüktağak E. B., Short M. R., Schwanemann R., Clark S., Ludvig D., Lynch K. M., Hargrove L., Tornero J., **Vianello, L.**, Pons J. L. *Simplifying Rehabilitation Control of Lower-Limb Exoskeletons in Five Ambulation Modes via Dataset-Driven State-Machine Calibration*. Journal of NeuroEngineering and Rehabilitation, 2026.
- Smith A. D., Ding W., **Vianello, L.**, Person E., Hauser K. *The EYESIGHT Robotic Eye Examination System for Nonmydriatic Indirect Retinal Imaging on Unconstrained Individuals*. Journal of Medical Robotics Research, 2026.
- **Vianello, L.**, Short M., Manczurowsky J., Küçüktağak E. B., Di Tommaso F., Nocco A., Bandini L., Clark S., Fiorenza A., Lunardini F., Canton A., Gandolla M., Pedrocchi A. L. G., Ambrosini E., Murie-Fernández M., Román C. B., Tornero J., Leon N., Sawers A., Patton J., Formica D., Tagliamonte N. L., Rauter G., Baur K., Just F., Hasson C. J., Novak V. D., Pons J. L. *Robot-Mediated Physical Human-Human Interaction in Rehabilitation: A Position Paper*. IEEE Reviews in Biomedical Engineering, 2026.
- Short M. R., Ludvig D., Di Tommaso F., **Vianello, L.**, Küçüktağak E. B., Perreault E. J., Hargrove L., Lynch K., Burdet E., Pons J. L. *Effects of Uni- and Bi-directional Interaction During Dyadic Ankle and Wrist Tracking*. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2025.
- Lhoste C., Küçüktağak E. B., **Vianello, L.**, Amato L., Short M. R., Lynch K. M., Pons J. L. *Deep-learning Estimation of Weight Distribution Using Joint Kinematics for Lower-Limb Exoskeleton Control*. IEEE Transactions on Medical Robotics and Bionics, 2024.
- **Vianello, L.**, Ivaldi S., Aubry A., Peternel L. *The Effects of Role Transitions and Adaptation in Human-Cobot Collaboration*. Journal of Intelligent Manufacturing, 2024.
- Short M. R., Ludvig D., Küçüktağak E. B., Wen Y., **Vianello, L.**, Perreault E. J., Hargrove L., Lynch K., Pons J. L. *Haptic Human-human Interaction During an Ankle Tracking Task: Effects of Virtual Connection Stiffness*. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2023.
- **Vianello, L.**, Gomes W., Stulp F., Aubry A., Maurice P., Ivaldi S. *Latent Ergonomics Maps: Real-time Visualization of Estimated Ergonomics of Human Movements*. Sensors, 2022.
- **Vianello, L.**, Penco L., Gomes W., You Y., Anzalone S. M., Maurice P., Thomas V., Ivaldi S. *Human-humanoid Interaction and Cooperation: A Review*. Current Robotics Reports, 2021.
- **Vianello, L.**, Mouret J. B., Dalin E., Aubry A., Ivaldi S. *Human Posture Prediction During Physical Human-robot Interaction*. IEEE Robotics and Automation Letters, 2021.

Conference Publications (Full Author Lists)

- Snyder G., **Vianello, L.**, Hargrove L., Elwin M. L., Pons J. *Learning Therapist Policy from Therapist-Exoskeleton-Patient Interaction*. IEEE International Conference on Robotics and Automation (ICRA), 2026.
- Embry K. R., **Vianello, L.**, Lipsey J., Ursetta F., Stephens M., Wang Z., Simon A. M., Ikeda A. J., Finucane S. B., Anarwala S., Hargrove L. J. *Proximal Powered Knee Placement: A Case Study*. IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2026.
- Amato L., **Vianello, L.**, Küçüktağak E. B., Lhoste C., Short M., Ludvig D., Lynch K., Hargrove L., Pons J. L. *Unidirectional Human-robot-human Physical Interaction for Gait Training*. IEEE International Conference on Biomedical Robotics and

Biomechatronics (BioRob), 2026.

- **Vianello, L.**, Gomes W., Maurice P., Aubry A., Ivaldi S. *Cooperation or Collaboration? On a Human-inspired Impedance Strategy in a Human-robot Co-manipulation Task*. IEEE International Conference on Robotics and Automation (ICRA), 2026.
- **Vianello, L.**, Lhoste C., Küçükçabak E. B., Short M. R., Hargrove L., Pons J. L. *Deep-learning Control of Lower-limb Exoskeletons via Simplified Therapist Input*. IEEE International Conference on Rehabilitation Robotics (ICORR), 2025.
- Cantón A., Lhoste C., **Vianello, L.**, Tornero J. *Development of a Customizable Assistance Algorithm for Lower-Limb Exoskeleton*. IEEE International Conference on Rehabilitation Robotics (ICORR), 2025.
- Sankar A., Küçükçabak E. B., Lhoste C., Short M., **Vianello, L.**, Pons J. *Action Diffusion for Learning Therapist Behavior in Exoskeleton-Assisted Dyadic Rehabilitation*. International Conference on NeuroRehabilitation (ICNR), 2024.
- Cantón A., Lhoste C., **Vianello, L.**, Tornero J., Pons J. L. *Development of a Customizable Assistance Algorithm for Lower-Limb Exoskeleton*. International Conference on NeuroRehabilitation (ICNR), 2024.
- Lhoste C., **Vianello, L.**, Cantón A., Küçükçabak E. B., Short M. R., Clark S. *Multiple Activities Rehabilitation Using Lower-Limb Exoskeletons: A Pilot Study with Two Stroke Patients*. International Conference on NeuroRehabilitation (ICNR), 2024.
- **Vianello, L.**, Lhoste C., Küçükçabak E. B., Short M. R., Pons J. L. *Locomotion Identification for Lower-Limb Exoskeletons*. International Conference on NeuroRehabilitation (ICNR), 2024.
- Küçükçabak E. B., Short M. R., **Vianello, L.**, Lhoste C., Lynch K. M., Pons J. L. *Exoskeleton-Mediated Physical Teacher-Student Interaction for Gait Training: A Pilot Study*. International Conference on NeuroRehabilitation (ICNR), 2024.
- Short M. R., Ludvig D., **Vianello, L.**, Di Tommaso F., Pons J. L. *Uni- and Bi-directional Haptic Human-Human Interaction During Upper- and Lower-Limb Tracking Tasks*. International Conference on NeuroRehabilitation (ICNR), 2024.
- **Vianello, L.**, Küçükçabak E. B., Short M., Lhoste C., Amato L., Lynch K., Pons J. *Exoskeleton-mediated Physical Human-human Interaction for a Sit-to-stand Rehabilitation Task*. IEEE International Conference on Robotics and Automation (ICRA), 2024.
- **Vianello, L.**, Mouret J. B., Dalin E., Aubry A., Ivaldi S. *Probabilistic Estimation of Postures During Human-robot Collaboration: An Ergonomics Perspective*. International Conference on Applied Human Factors and Ergonomics (AHFE), 2021.

Activities

2026	Special Session organizer , International conference in Neuro-Rehabilitation (ICNR)	Korea
2025	Workshop Organizer , International Consortium for Rehabilitation Robotics (ICORR)	Chicago
2025	Special Issue Guest editor , Journal of NeuroEngineering and Rehabilitation	
2025	Special Issue Guest editor , Journal of Wearable Technologies	
2024	Special Session organizer , International conference in Neuro-Rehabilitation (ICNR)	Spain